

Disproof of conjecture 00000003485

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Verdict: FALSE

We disprove conjecture 00000003485 as stated. The refutation is unconditional and elementary, and the witness is the smallest possible: the directed 5-cycle. It has out-degree 1 everywhere, directed girth 5, and yet no kernel. The same failure occurs for every directed odd cycle, and the failure is stable under strongly connected blow-ups, so it is not an artefact of out-degree 1.

1 The conjecture

Quoted verbatim from `conjectures/00000003485.md`:

Definition: A kernel of a digraph is an independent out-stable set, with the existence criterion the core problem. **Conjecture:** Digraphs with bounded out-degree and girth at least five have kernels; the critical configuration of the condition is the directed odd cycle with a one-point source, and the bound is a tightening of Richardson’s theorem. (kernel existence strengthened condition)

定义: 有向图的核为独立外稳定集, 其存在性判据是核心问题。 **猜想:** 出度有界且围长至少五的有向图存在核; 条件的临界构型为有向奇圈加单点源且该界为 Richardson 定理的紧化。(核存在强化条件)

Definition 1 (digraph). A digraph $D = (V, A)$ consists of a vertex set V and an arc set $A \subseteq V \times V$.

Definition 2 (kernel). A *kernel* of $D = (V, A)$ is a set $S \subseteq V$ that is

1. *independent*: no arc joins two members of S in either direction, i.e. for all $u, v \in S$ neither $(u, v) \in A$ nor $(v, u) \in A$; and
2. *out-stable* (absorbing): every vertex outside S has an arc into S , i.e. for every $v \in V \setminus S$ there is $s \in S$ with $(v, s) \in A$.

Definition 3 (out-degree, girth). The *out-degree* of v is $d^+(v) = |\{w : (v, w) \in A\}|$. The *directed girth* is the length of the shortest directed cycle, i.e. the least $k \geq 1$ with a closed directed walk of length k ; it is $+\infty$ for an acyclic digraph. The (*underlying undirected*) *girth* is the length of the shortest cycle in the underlying undirected graph.

The hypothesis “bounded out-degree” is read in its weakest useful form, as a global bound $d^+(v) \leq B$ for a fixed constant B . A universal implication of the form “every digraph with bounded out-degree and girth ≥ 5 has a kernel” is refuted by a single digraph satisfying both hypotheses and having no kernel.

2 The witness: the directed 5-cycle

Definition 4 (C_5). Let C_5 be the digraph with vertex set $V = \{0, 1, 2, 3, 4\}$ and arc set

$$A = \{(i, i + 1 \bmod 5) : i \in V\}.$$

It is the directed cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$.

Lemma 5 (hypotheses hold). *The digraph C_5 has out-degree 1 at every vertex, hence bounded out-degree, and both its directed girth and its underlying undirected girth are equal to $5 \geq 5$.*

Proof. Every vertex has exactly one out-neighbour, its cyclic successor, so $d^+(v) = 1$ for all v ; in particular the out-degree is bounded (take $B = 1$). The directed closed walk $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$ has length 5, so the directed girth is at most 5. There is no closed directed walk of length 1, 2, 3, or 4: every arc moves the index up by exactly 1 modulo 5, so a closed walk of length k would force $k \equiv 0 \pmod{5}$, and no positive $k < 5$ satisfies this. Hence the directed girth is exactly 5. The underlying undirected graph is the ordinary cycle $0-1-2-3-4-0$, whose girth is 5. $\square$

3 C_5 has no kernel

Theorem 6. *The directed 5-cycle C_5 has no kernel.*

Proof. We use two elementary facts about independence and absorption in C_5 .

Independence. An independent set contains no two cyclically consecutive vertices, because $(i, i+1) \in A$ for every i . Hence an independent set is an independent set of the underlying undirected 5-cycle, whose independence number is 2; so $|S| \leq 2$, and if $|S| = 2$ then $S = \{i, i+2\}$ for some i (the only non-adjacent pairs are the pairs at cyclic distance 2).

Absorption. If $v \notin S$, the unique arc leaving v is $(v, v+1)$, so v can have an arc into S only if $v+1 \in S$. Thus out-stability reads:

$$\text{for every } v \in V : \quad v \in S \text{ or } v+1 \in S. \tag{1}$$

Now let S be independent, and consider its size.

- $|S| = 0$: then $0 \notin S$ and $0+1 = 1 \notin S$, violating (1).
- $|S| = 1$, say $S = \{i\}$: the vertex $i+2$ lies outside S and its successor $i+3$ is not i , so $i+3 \notin S$; (1) fails at $v = i+2$.
- $|S| = 2$, say $S = \{i, i+2\}$: the vertex $i+3$ lies outside S and its successor $i+4$ is not i or $i+2$, so $i+4 \notin S$; (1) fails at $v = i+3$.
- $|S| \geq 3$: impossible, since the independence number is 2.

Every candidate fails, so no subset $S \subseteq V$ is both independent and out-stable. Hence C_5 has no kernel. $\square$

Independent check by exhaustive enumeration. There are $2^5 = 32$ subsets of V . Exactly 11 of them are independent: $\emptyset$; the five singletons $\{0\}, \{1\}, \{2\}, \{3\}, \{4\}$; and the five non-adjacent pairs $\{0, 2\}, \{1, 3\}, \{2, 4\}, \{0, 3\}, \{1, 4\}$. For each one, the table below exhibits a vertex that lies outside S and has no arc into S , so none is out-stable; the remaining 21 subsets are already not independent.

independent set S	a vertex outside S with no arc into S
$\emptyset$	0 (its arc goes to $1 \notin S$)
$\{0\}$	2 (its arc goes to $3 \notin S$)
$\{1\}$	3 (its arc goes to $4 \notin S$)
$\{2\}$	4 (its arc goes to $0 \notin S$)
$\{3\}$	0 (its arc goes to $1 \notin S$)
$\{4\}$	1 (its arc goes to $2 \notin S$)
$\{0, 2\}$	3 (its arc goes to $4 \notin S$)
$\{1, 3\}$	4 (its arc goes to $0 \notin S$)
$\{2, 4\}$	0 (its arc goes to $1 \notin S$)
$\{0, 3\}$	1 (its arc goes to $2 \notin S$)
$\{1, 4\}$	2 (its arc goes to $3 \notin S$)

Thus a brute force over all 32 subsets finds 0 kernels, confirming Theorem 6. The computation is reproduced by `reproduce.py` and formalised in `lean4/Main.lean`. $\square$

The case analysis above is also the general phenomenon, which we record next.

4 The general odd-cycle obstruction

Theorem 7. *For $n \geq 3$, the directed n -cycle C_n has a kernel if and only if n is even. When n is even it has exactly two kernels, the even-indexed and the odd-indexed vertices.*

Proof. Write the vertices as $\mathbb{Z}/n$ with arcs $i \rightarrow i+1$. Independence forbids two cyclically consecutive vertices from both lying in S ; out-stability (1) forbids two cyclically consecutive vertices from both lying outside S . Hence, for each i , exactly one of i and $i+1$ lies in S . Starting from $i=0$, this forces

$$i \in S \iff i \equiv \varepsilon \pmod{2}$$

for a fixed $\varepsilon \in \{0, 1\}$. Consistency around the cycle requires $n \equiv 0 \pmod{2}$, and then the two choices of ε give the two stated kernels. For odd n no set S satisfies both conditions, so C_n has no kernel. $\square$

Corollary 8. *C_5 , C_7 , and C_9 have no kernels; in particular the conjecture fails for every odd $n \geq 3$, with $n=5$ the smallest value compatible with the conjectured girth bound.*

Remark 9 (Richardson’s theorem). Richardson’s theorem states that every digraph with no odd directed cycle has a kernel. The conjectured condition “girth at least five” does not imply the absence of odd directed cycles: the directed 5-cycle is itself an odd directed cycle and has girth 5. The condition is therefore not a valid strengthening of Richardson’s theorem, and C_5 shows this directly.

5 Robustness: blow-ups, and either reading of girth

Fix $m \geq 1$. The m -blow-up of C_5 is obtained by replacing each vertex i by an independent part $P_i = \{(i, k) : 0 \leq k < m\}$ of size m and putting an arc from every vertex of P_i to every vertex of $P_{i+1 \bmod 5}$. For $m=1$ this is exactly C_5 .

Theorem 10. *For every $m \geq 1$, the m -blow-up of C_5 has out-degree m , directed girth 5, and no kernel.*

Proof. Every vertex of a part P_i has arcs to all m vertices of P_{i+1} , and to no other part, so the out-degree is exactly m . Every directed cycle projects to a closed walk in C_5 that advances the part index by 1 at each step, so its length is a positive multiple of 5; the cycle through $P_0, \dots, P_4$ has length 5. Hence the directed girth is 5.

For the kernel, write $S_i = S \cap P_i$ and let $b_i = 1$ if $S_i \neq \emptyset$ and $b_i = 0$ otherwise. If $b_i = b_{i+1} = 1$ then picking $u \in S_i$, $v \in S_{i+1}$ gives the arc (u, v) , contradicting independence; so no two consecutive b 's are both 1. If $b_{i+1} = 0$ then, whenever some vertex of P_i lies outside S (that is, $S_i \neq P_i$), that vertex has all its out-arcs landing in P_{i+1} and hence has no arc into S , contradicting out-stability. Consequently

$$b_{i+1} = 0 \implies S_i = P_i. \quad (2)$$

If all $b_i = 0$, then $S_0 = \emptyset \neq P_0$ while $b_1 = 0$, contradicting (2). So some $b_i = 1$. Independence then forces $b_{i-1} = b_{i+1} = 0$. Apply (2) at $i + 1$: $b_{i+1} = 0$ gives $S_{i+1} = P_{i+1} \neq \emptyset$, so $b_{i+2} = 1$. Continuing around the cycle, the b 's alternate $1, 0, 1, 0, \dots$, which is consistent only if the number of parts is even; but there are 5 parts. Equivalently, $b_{i+5} = b_i$ would be forced to be both 1 and 0. Hence no S is both independent and out-stable. $\square$

Remark 11 (robustness). Theorem 10 shows the failure is not an artefact of out-degree 1: every out-degree bound m admits a kernel-free digraph of girth 5. Since the underlying undirected graph of C_5 is a 5-cycle whose girth is 5 as well, the counterexample also refutes the conjecture under the undirected reading of “girth”. Finally, the failure is unconditional in the sense that it uses no asymptotic or numerical coincidence: it is a structural parity obstruction.

6 Reproducibility

`reproduce.py` (pure Python 3, standard library only) brute-forces all 2^n subsets of the directed n -cycle for $n = 3, 5, 7, 9$, confirms that the number of kernels is 0 in every case, verifies that C_5 has out-degree 1 and directed and undirected girth 5, and repeats the check for the blow-ups of C_5 with part sizes $m = 1, 2, 3$. It prints `PASS` and exits with status 0 when every fact verifies, and exits non-zero otherwise.

The document is built with `tectonic main.tex`; the PDF is `build/main.pdf`.

7 Formalisation

The refutation is formalised in Lean 4 in `lean4/`, using core Lean only (`import Std`, no `Mathlib`, no `sorry`). The digraph is the decidable arc predicate `arc5 i j := decide (j = i + 1)` on `Fin 5`; subsets are indicator functions `Fin 5 → Bool`; and the predicates `independent5`, `outStable5`, `kernel5` are `Bool`-valued. The main theorem

$$\text{no_kernel5} : \forall S : \text{Fin } 5 \rightarrow \text{Bool}, \text{kernel5 } S = \text{false}$$

is proved by case analysis on the five values of S — an exhaustive split of the 32 subsets — with each closed case discharged by `decide`. The out-degree and girth facts are the theorems `outdeg5_le_one`, `girth5_ge_five`, and `girth5_le_five`; the explicit 32-subset brute force is `kernelMasks5_nil`;

and the odd cycles C_3 , C_7 are covered by `no_kernel13`, `no_kernel17`. The combined statement `conjecture_00000003485_false` asserts that the hypotheses of the conjecture hold for C_5 while no kernel exists.

8 Status against the submission rules

- **LaTeX source** — present (`main.tex`), a standalone article using only `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, and `parskip`, designed to compile with `tectonic main.tex`.
- **PDF document** — `build/main.pdf`, produced by `tectonic main.tex`.
- **Lean 4 project** — present under `lean4/`, with `lean-toolchain` (pinned to `leanprover/lean4:v4.33.1`), `lakefile.toml` (library `Main`, name `tlmc3485`), `Main.lean`, `Check.lean`, and `README.md`. It contains no `sorry` and no `Mathlib` dependency; `Check.lean` reports the axioms of every theorem, and none reports `sorryAx`.
- **Reproduction** — `reproduce.py` is dependency-free and verifies all claims.